Algorithm 1: Gate-based ensemble of TSFMs for few-shot fault prognosis

Input : window stream $\{X_t\}$; target y; few-shot ratio ρ; pretrained TSFMs $\{\theta_0^{(i)}\}$
Output: fused forecasts $\hat{y}^{\text{gate}}$ and alarm decisions $\{A_t\}$ on the test split

Fit control limits (ℓ, u) and scaler on healthy training data ▷ Eq. (3), (9)
Draw nested few-shot subset $\mathcal{S}_\rho$ by seeded permutation ▷ Eq. (8)
for *each operator* $i \in$ {Timer-XL-raw, Timer-XL-diff, Time-MoE, Sundial} **do**
  Obtain $\theta^{(i)}$: few-shot align Timer-XL views from $\theta_0^{(i)}$, else use pretrained ▷ Eq. (7)
  Infer $\hat{y}^{(i)}$ on validation and test (diff view restored) ▷ Eq. (10), (11)
Extract context features $\varphi(X)$ on validation and test ▷ Eq. (12)
Train gate g_ψ on validation with operators frozen ▷ Eq. (13), (19)
Fuse operators on test by per-horizon routing ▷ Eq. (14)
Map fused forecast to alarms, lead times and preFAR ▷ Eq. (15)–(18)
return $\hat{y}^{\text{gate}}$, $\{A_t\}$